

Tan Jun Zuo

junzuo100@gmail.com | (+65) 85116801 | <https://www.linkedin.com/in/tan-jun-zuo-84409810b/>

PROFESSIONAL SUMMARY

AI/ML Engineer and Data Scientist with over 4 years of experience developing machine learning models and AI-powered applications within the financial services industry. Experienced across the full ML lifecycle, including data exploration, feature engineering, model development, explainability, and deployment. Skilled in building production-ready AI solutions using LLMs, Retrieval-Augmented Generation (RAG), vector databases, and workflow automation, with a passion for delivering scalable, intelligent systems that solve real-world business problems

TECHNICAL SKILLS

- **Programming & ML:** Python, Object-Oriented Programming (OOP), SQL, Git, Pandas, NumPy, Scikit-learn, LGBM, PyTorch, TensorFlow, Keras, XGBoost, LightGBM
- **Data & Cloud:** Azure, Databricks, Spark, Snowflake
- **NLP & GenAI:** Hugging Face, LangChain, Retrieval-Augmented Generation (RAG), Prompt Engineering, AI Agents, MCP (Model Context Protocol), n8n Workflow Automation, LLM Integration, Transformers
- **Techniques:** CNN, Logistic Regression, Credit Scoring Models, ECL Modeling, Macroeconomic Forecasting, Transformers, Deep Learning, Time Series Forecasting, Feature Engineering, Model Deployment, Machine Learning Pipelines
- **Tools & Platforms:** Flask, FastAPI, Docker, Power BI, Tableau, Git/GitHub, Azure DevOps
- **Methodologies:** Agile, Scrum, EDA, Supervised & Unsupervised Learning
- **Certifications:**
 - **Quantexa Certified Data Engineer:** Specialized in entity resolution and network generation, enabling the discovery of hidden relationships across millions of records.
 - **Heicoders GA100: Generative AI for Automation & Productivity:** Built AI-powered workflow automations using LLMs, AI agents, MCP, APIs, and n8n to streamline business processes.

PROFESSIONAL EXPERIENCE

Synpulse Singapore Pte. Ltd.

Data Scientist | Nov 2025 – Present

- Played a key role in a **Trade-Based Money Laundering (TBML)** detection enhancement project, designing new red flag indicators and ML models to improve risk coverage.
- Developed transactional and network-based (**Quantexa**) indicators, leveraging entity resolution and link analysis to uncover hidden risk relationships.
- Built a rules-augmented outlier detection model (LOF + logistic regression) on 194K customers, combining rule-based and ML approaches.
- Achieved 40% reduction in alerts and **60% reduction in false positives**, while maintaining **87.5% recall** on suspicious cases.
- Identified 149 new high-risk customers missed by the incumbent system, expanding detection capability.

Corestrat Capital Pte Ltd – Singapore

Data Scientist | Jun 2022 – Nov 2025

- Developed a **credit scorecard** to evaluate loan applicants and set optimal interest rates using logistic regression, reducing default rates by 15% and improving loan pricing accuracy.
- Primary **backend developer and data engineering lead** for Corestrat's flagship product **Intellidecision.ai**, a one-stop predictive model builder platform enabling model creation and strategy simulation in under 1 hour. Designed and implemented scalable Python-based data science APIs for automated cleaning, transformation, clustering, and multi-model training - reducing decision-making time from days to under 1 hour.
- Built an **Expected Credit Loss (ECL) model** with macroeconomic forecasting, improving credit risk prediction accuracy by 15% and ensuring compliance with IFRS 9 standards.
- Engineered a **cloud-based architecture** for an automated options trading platform, enabling end-to-end trade execution from real-time market data ingestion to automated buy/sell order placement. Integrated Interactive Brokers (IBKR) APIs with a containerized iBeam authentication service on Azure Container Instances and deployed Azure Function Apps for signal generation and execution, laying the groundwork for fully autonomous trading strategies.

Home Team Science & Technology Agency – Singapore

Data Scientist Intern | Jan 2021 – May 2021

- Applied **NLP** techniques to extract insights from unstructured incident reports. Boosted actionable intelligence by 30%.
 - Built **clustering** pipelines to correlate related cases. Achieved 86% match accuracy in linking incidents.
-

KEY PROJECTS

- **Enterprise AI Customer Service Automation Platform (GA100 Capstone Project):** Built a production-style **Retrieval-Augmented Generation (RAG)** customer support platform using **n8n, OpenAI GPT-5, Pinecone, Gmail, and Google Drive**, automating the end-to-end customer support workflow from document ingestion and vectorization to AI-powered email response generation. Implemented automated knowledge synchronization, semantic search over enterprise documentation, prompt injection safeguards, retrieval-only response generation to minimize hallucinations, and standardized customer communication templates, demonstrating the design and deployment of secure, enterprise-grade Generative AI workflows.
 - **Financial RAG Application** – Developed a **retrieval-augmented generation (RAG)** system using **Hugging Face models** to enable context-aware Q&A on financial reports. Implemented document loaders, vector storage, and RAG chain logic in Python, applying semantic search and embeddings to extract insights from real financial data.
 - **Deep Learning for Tree Defect Detection** - Developed a **CNN-based defect detection system** to identify internal decay in trees, mitigating risks of tree-fall incidents. Processed scanning signal data in MATLAB to reconstruct internal tree structures and trained a U-Net model to segment defects.
-

EDUCATION

Nanyang Technological University (NTU) – Singapore

Bachelor of Engineering (Electrical & Electronic Engineering) | Aug 2019 – May 2022

- Specialization: **Infocommunications & Machine Learning**